Interactive News Website UI with Embedded Policy Query Agent

Technical Architecture & Implementation Specification • SingGov PolicyWire • System Integration

Target Framework: React 19 + Vite + Tailwind CSS	**Backend:** FastAPI / Bedrock Nova Lite	**Version:** 2.4.0 (Unified Release)

1. Goal Description

Build a modern, highly polished news-portal styled user interface for Singapore public policies. On the frontpage, users browse recent policy announcements organized like an authoritative editorial news publication (newspaper masthead, breaking ticker, lead story hero, category desks, trending policy updates). When a user clicks on any policy article, the central article reading body is dynamically replaced by a dedicated, interactive **Policy Query Agent chatbot** stream with an auto-generated executive briefing, instant suggested inquiry pills, and real-time grounded conversational Q&A powered by official Singapore Government gazettes and circulars.

2. Critical Architecture Decisions (User Review)

CRITICAL ARCHITECTURAL HIGHLIGHTS & GUARDRAILS

• **Editorial Theme & Branding:** Styled as *'The Singapore Policy Herald / PolicyWire SG'*, combining the authority of top broadsheets (Straits Times, FT, Reuters, CNA) with sleek dark/light mode responsive tokens.

• **Ground-Truth Policy Registry:** Built a comprehensive, type-safe policy dataset covering key 2024–2026 national frameworks: BTO Plus/Prime Housing Framework, CPF SA Closure & ERS Adjustments, MediShield Life Claim Revisions, SkillsFuture Level-Up, Shared Parental Leave 2026, and the Majulah Package.

• **Seamless Query Agent Priming:** Opening any policy article primes the embedded Query Agent with the policy's structured context and official government sources, generating an immediate context briefing and interactive question-answering session.

• **No-Embedding & Direct Citation Policy:** In strict compliance with system architecture rules, queries avoid vector databases, relying on live multi-source web scraping and document context feeding directly to Amazon Bedrock Nova Lite.

3. Proposed Changes & File Breakdown

The platform is structured into modular components within the frontend application and interconnected backend endpoints:

Type	File Path	Functional Responsibility
NEW	`src/data/policies.ts`	Central policy registry with metadata, official .gov.sg sources, key metrics (income ceilings, payout caps), and preset briefing templates.
NEW	`src/components/NewsHeader.tsx`	Newspaper masthead with live date stamp, breaking ticker, navigation tabs, and Parliamentary Debates portal link.
NEW	`src/components/PolicyHeroCard.tsx`	Frontpage editorial lead story card with ministry badges, publication timestamp, lead abstract, and 'Read & Consult AI' trigger.
NEW	`src/components/PolicyNewsCard.tsx`	Responsive multi-column news card with category indicator, headline, summary snippet, and interactive hover states.
NEW	`src/pages/PolicyArticlePage.tsx`	Interactive article view replacing static text with the embedded Query Agent chat stream, official sources sidebar, and key fact pills.
NEW	`src/pages/PolicyNewsFeed.tsx`	Editorial frontpage feed displaying breaking alerts, hero banner, category desks (Housing, CPF, Health), and trending queries.
MOD	`src/App.tsx`	Unified top-level routing container coordinating header navigation, global state, and embedded policy assistant views.
MOD	`src/main.tsx`	React Router DOM configuration for /, /policy/:id, /debates, and /query.
MOD	`src/index.css`	Refined typographic hierarchy (Merriweather / Outfit / Inter), newspaper border rules, and animated chat response bubbles.

4. Policy Ground-Truth & Parameter Enforcement

To eliminate LLM hallucinations on critical statutory limits, exact statutory thresholds were encoded into the agent's baseline knowledge repository:

Policy Scheme	Category	Enforced Statutory Parameters & Thresholds	Primary Official Source
BTO Plus/Prime Framework	Housing (HDB)	**Singles:** $8,000/mo income ceiling **Couples/Families:** $16,000/mo income ceiling **MOP:** 10 years; Subsidy Clawback applies	HDB Gazetted Guidelines (Oct 2024)
CPF SA Closure & ERS	CPF & Retirement	**Closure:** Age 55 SA balances transferred to RA up to FRS, excess to OA. **ERS:** 4x BRS ($426,000 in 2025)	CPF Board Circular (Budget 2024)
Shared Parental Leave	Family & Grants	**Total:** 10 weeks shared leave by 1 Apr 2026 (6 weeks from 1 Apr 2025). Paid by Govt up to $2,500/week.	MSF & MOM Joint Release
SkillsFuture Level-Up	Manpower (MOM)	**Top-up:** $4,000 for Singaporeans aged 40+. **Training Allowance:** Up to $3,000/month for up to 24 months.	SkillsFuture Singapore (SSG)

5. Verification & Validation Plan

- **Frontend Build & Typecheck:** Executed npm run build in query_agent/frontend with zero TypeScript or JSX bundling errors.
- **Navigation & Routing Verification:** Verified routes /, /policy/:id, /debates, and /query resolve smoothly with correct breadcrumbs and page titles.
- **Embedded Query Agent Interaction:** Validated that clicking a policy loads the executive brief and quick-ask chips, and queries to POST /query return structured, cited responses.
- **Accuracy & Hallucination Testing:** Conducted live queries against HDB income ceilings, CPF retirement sums, and maternity leave rules to ensure strict adherence to official figures.
- **Backend Integration & Proxying:** Verified that Vite proxy routes /api/debates/* and /debates/* to the BlogPost service (port 8001) while /query routes to the QueryAgent service (port 8000).

Architecture Status: APPROVED & IMPLEMENTED **Audit Date:** September 2026 **Target Environment:** Production / Staging